

The Impact of Virtual Agent Locomotion and Postures on Social Perception and Brain Activity in MR

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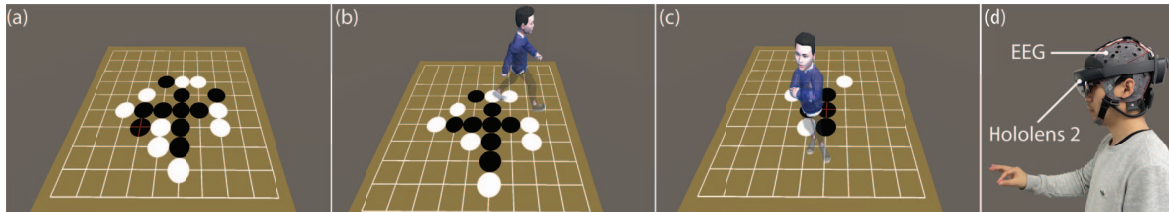


Figure 1: Virtual agent conditions and hardware: a) Speech only agent, b) embodied agent with both Speech and Locomotion, c) embodied agent with Speech, Locomotion, and Postures, and d) user wearing an EEG headset and Hololens2 playing against the agent.

ABSTRACT

When people interact with virtual characters, the character's non-verbal cues can change the user's perception of and engagement with it. In this work, we investigate the impact of virtual agent locomotion and emotional posture on user engagement, as measured by subjective questionnaires and EEG signals. We found the EEG-based engagement index was significantly lower in the postural agent condition, which was opposite to the questionnaire-based engagement result. From these results, we differentiate social engagement from task engagement and discuss their influence on overall perceived engagement.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

Unlike traditional Intelligent Virtual Agents (IVAs) presented on a two-dimensional screen, Mixed Reality Agents (MiRAs) are embodied in the same physical space as real users. So MiRAs can move around and present postures to communicate with the users [4]. Exploring the impact of virtual agent locomotion and postures in an MR environment on human perception and behaviour is important for the design of MiRAs.

The human brain drives all cognitive activities like attention and memory. The electrical signals of brain activity can be captured by Electroencephalography (EEG), which has been used for measuring cognitive load [1], engagement [8] and so on. However, it is unclear how the virtual agent nonverbal cues like postures and locomotion in MR space can influence user engagement, as measured by EEG.

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In this work, we conduct a within-subjects user study to explore how the MiRAs' locomotion and postures influence the user's brain activity and feelings of social presence and engagement. Our results suggested that a MiRA using postures improved social engagement as measured by questionnaires, while reducing the overall engagement as measured by EEG. Based on these findings, we suggest that it is important to differentiate social engagement, task engagement and overall engagement while designing the MiRAs' communication cues.

2 METHOD

2.1 Experimental Design

To explore the effects of the MiRAs' locomotion and postures on social presence and engagement, a within-subjects study was designed where participants played an Gobang game¹ in Mixed Reality (MR). There were three different agent conditions: **S**: As a baseline, the virtual agent had no visual representation and used speech only (See figure 1 a)). **S+L**: Based on the S configuration, this agent also had a virtual human body walking on top of the Gobang board to place pieces (See figure 1 b)). **S+L+P**: Based on the S+L condition, the agent also used postures to express thinking behavior and emotions based on the game state (See figure 1 c)).

2.2 Procedure

In the beginning, participants were introduced on the procedure and content of the experiment. After filling out a consent form and a demographic questionnaire, they were trained by playing the MR Gobang game against the computer without any agent. Once participants got familiar with the game rules and MR interaction, the actual experiment and data collection started. Each participant took three trials in total and the whole experiment lasted around 80 minutes. After the experiment, a structured interview was conducted. The three trial conditions were experienced in a Latin square sequence to reduce any learning effects.

2.3 Measures

We collected users' EEG signals, behavioural data as well as subjective feedback. From the EEG signals, we calculated an EEG-based

¹<https://en.wikipedia.org/wiki/Gomoku>

engagement index using the method from [8]. For the behavioural data, we logged user eye gaze, task performance and average speed. With the logged eye gaze data, we further calculated the time duration of gazing at the virtual human. The task performance was measured by counting the number of times the user won, lost, or tied in the Gobang game. The average speed was calculated to reflect players' movement speed while playing the pieces in each condition.

For the subjective measures, we used the social presence questionnaire [2], and the engagement questionnaire [6] to understand users' feelings about interacting with the virtual agent. We also used the Self-Assessment Manikin questionnaire (SAM) [5] to measure user emotional arousal and valence during the game, and the NASA-TLX questionnaire [3] to measure user workload.

3 RESULTS AND DISCUSSION

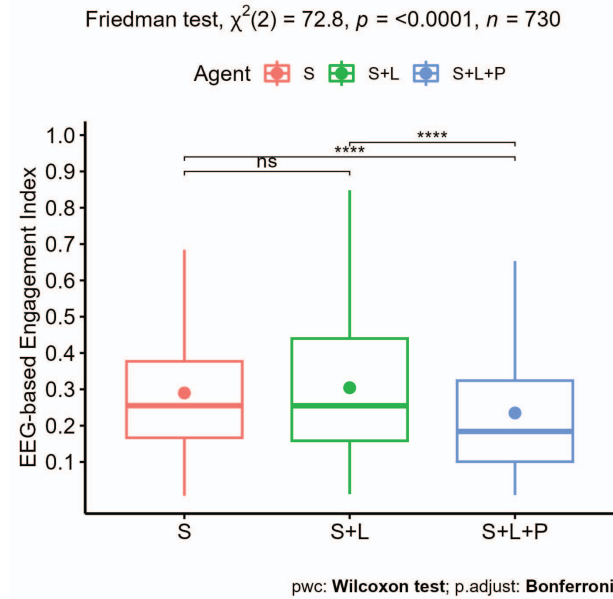


Figure 2: Results of the EEG-based engagement index. Statistical significance: **** ($p < 0.0001$).

We collected data from 16 participants (10 male 6 female) with their ages ranging from 20 to 33 years old ($M = 27.6, SD = 3.6$). The EEG engagement index results showed that the S+L+P condition had significantly lower engagement than the other two conditions (See figure 2). However, The engagement questionnaire indicated that the S+L+P condition had significantly higher engagement compared with other two conditions (See figure 3). The S+L+P condition was also rated significantly higher value in the social presence and SAM arousal questionnaires. We could not see any significance in other measures shown in section 2.3.

Since the EEG captures users' brain activity, the EEG-based engagement index reflected overall engagement level including both social engagement and task engagement. However, the engagement questionnaire used by our study mainly focuses on social engagement [7] because each statement is related to interaction with the virtual agent. We conclude that 1) Providing a virtual human body with locomotion made no difference from playing against the speech-only agent on engagement and social presence, and 2) adding postures to the embodied agent improved social engagement while reducing the overall engagement.

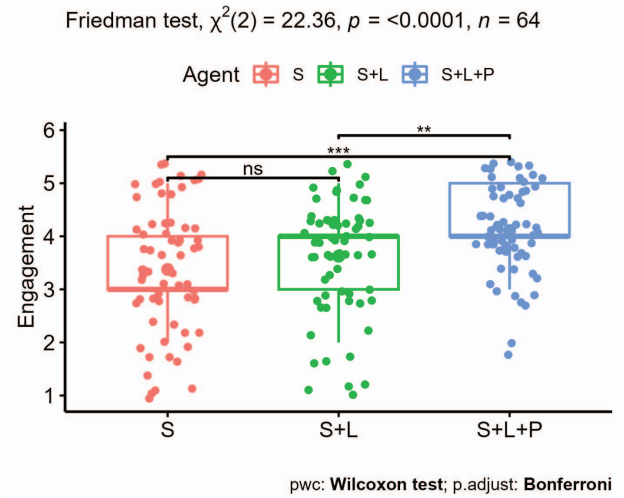


Figure 3: Results of the engagement questionnaire. (5-point Likert Scale from 1 to 5; Statistical significance: ** ($p < 0.01$), *** ($p < 0.001$)).

4 CONCLUSION

In this paper, we explored the impact of mixed reality agents' non-verbal cues like locomotion and emotional postures on perceived social presence, engagement, and user brain activity. We presented a with-subject user study where participants were asked to win against the mixed reality virtual agent over a Gobang game. Our results showed that the EEG-based engagement index was lowest in the postural agent condition, which was opposite to the result of the engagement questionnaire. This demonstrates that needed to differentiate task engagement and social engagement by combining behaviors, questionnaires, and physiological sensors like EEG in human-agent interaction. In the future we plan to explore how would the virtual agent's locomotion and postures influence user perception when it acts as an assistant rather than opponent in the Gobang game.

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